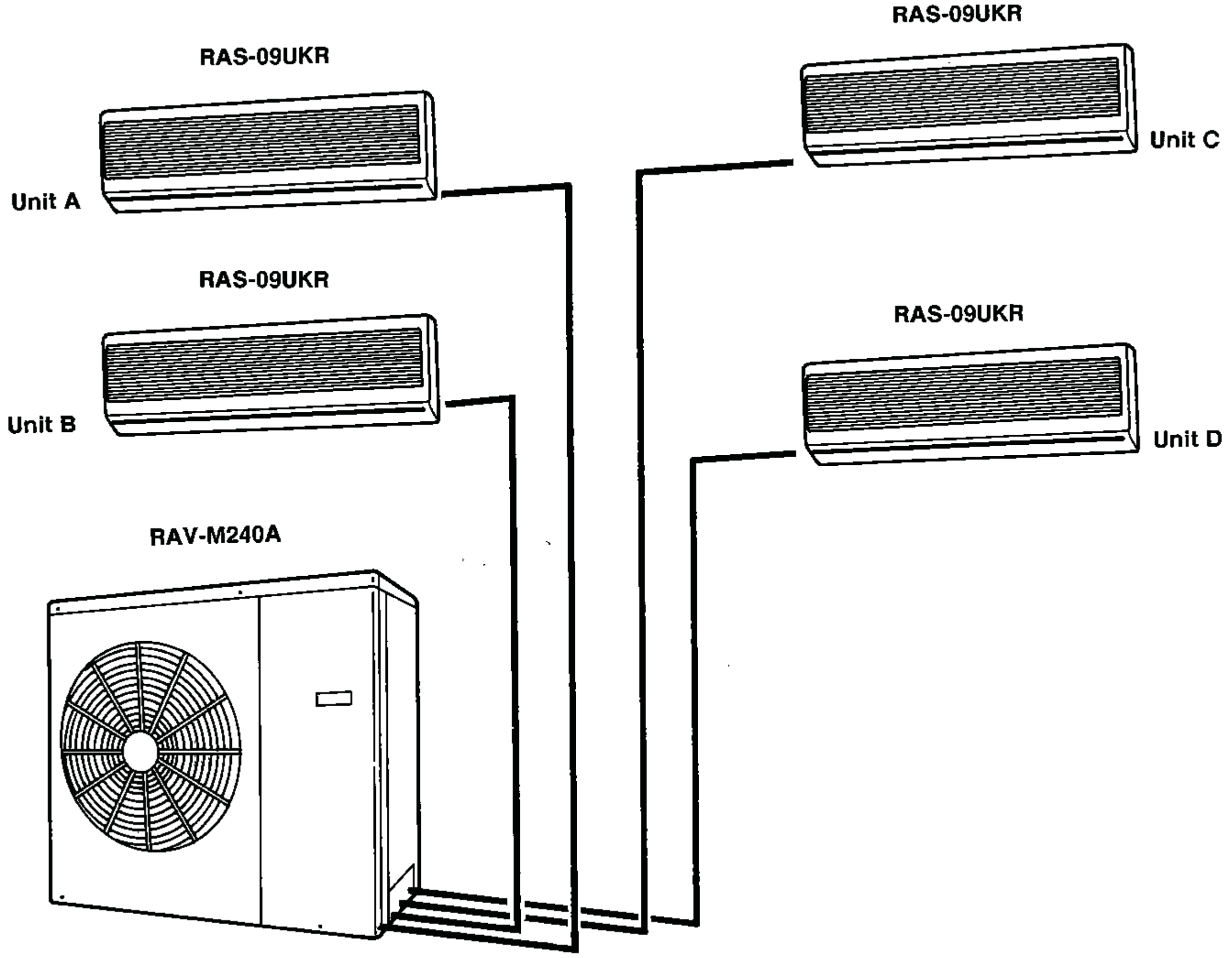
SERVICE DATA
FILE NO. 300-974

TOSHIBA

ROOM AIR-CONDITIONER

RAV-M240A (Combination with RAS-09UKR)



Specifications are subject to change without notice.

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CONTENTS

1. SPECIFICATIONS

ITEM		MODEL	RAV-M240A	
Operation		Unit	2 indoor unit A or B / C or D	4 indoor unit A / B / C / D
	*1	BTU/h	12000	6000
Cooling capacity		kcal/h	6000	6000
	*1	BTU/h	24000	24000
		kW	6.9	6.9
Power source		Phase	1	
		V	220-240	
		Hz	50	
Power consumption	Cooling	kW	1.86	2.45
Power factor	Cooling	%	97.4	97
Running current	Cooling	A	8.3	11.0
Starting current	Cooling	A	58	
Operating noise (SPL)		dB(A)	53	
Refrigerant	Name of refrigerant		R-22	
	Charge volume	kg	0.9 + 0.9	
Refrigerant control			Capillary tube	
Interconnection pipe	Gas side size	mm (in.)	9.5 (3/8")	
	Coupler style		Flare	
	Liquid side size	mm (in.)	6.4 (1/4")	
	Coupler style		Flare	
	Standard length	m (ft)	7.6 (25)	
	Maximum length *2 (of one way)	m (ft)	A + B < 15m, A – B < 8m A or B < 10m	
	Minimum length	m (ft)	2 (7)	
Maximum height	Indoor unit higher	m (ft)	5 (16)	
	Outdoor unit higher	m (ft)	5 (16)	
INDOOR UNIT Model			**RAS-09UKR**, SERVICE DATA FILE NO. 300-900	
OUTDOOR UNIT Model			**RAV-M240A**	
Dimensions	Height	mm (ft-in.)	790 (2'7-7/64")	
	Width	mm (ft-in.)	880 (34.64")	
	Depth	mm (ft-in.)	310 (12.2")	
Net weight		kg (lbs)	70 (154.3)	
Condenser type			Finned tube	
Condenser fan type			Propeller fan	
Fan motor output		W	63	
Compressor	Model		PH160X2-4L	
	Output	W	1100	
Protective device			Fuse Inner overload relay	

Specifications are subject to change without notice.

Note 1: Cooling capacity is based on the following temperature conditions. [Condition A]
Evaporator air inlet temperature: 27°C DB (80°F DB)
19.5°C WB (67°F WB)
Condenser air inlet temperature: 35°C DB (95°F DB)

Note 2: These mean equivalent length.

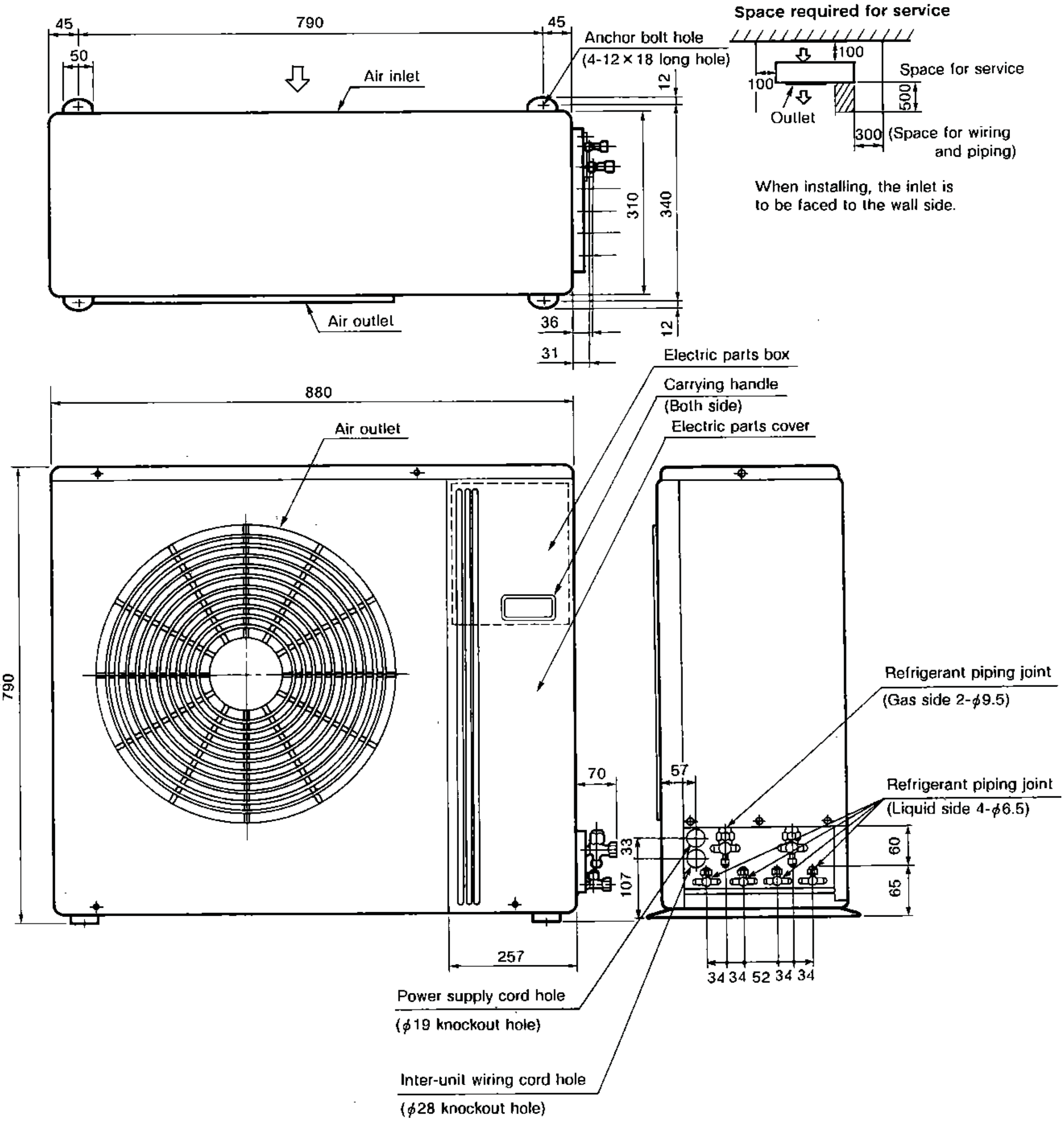
Note 3: Operating range of the units
Evaporator air inlet temperature
Maximum 32°C DB, 22.5°C WB (95°F DB, 73°F WB)
Minimum 21°C DB, 15.5°C WB (70°F DB, 60°F WB)

Condenser air inlet temperature
Maximum 43°C DB (109°F DB)
Minimum 21°C DB (70°F DB)

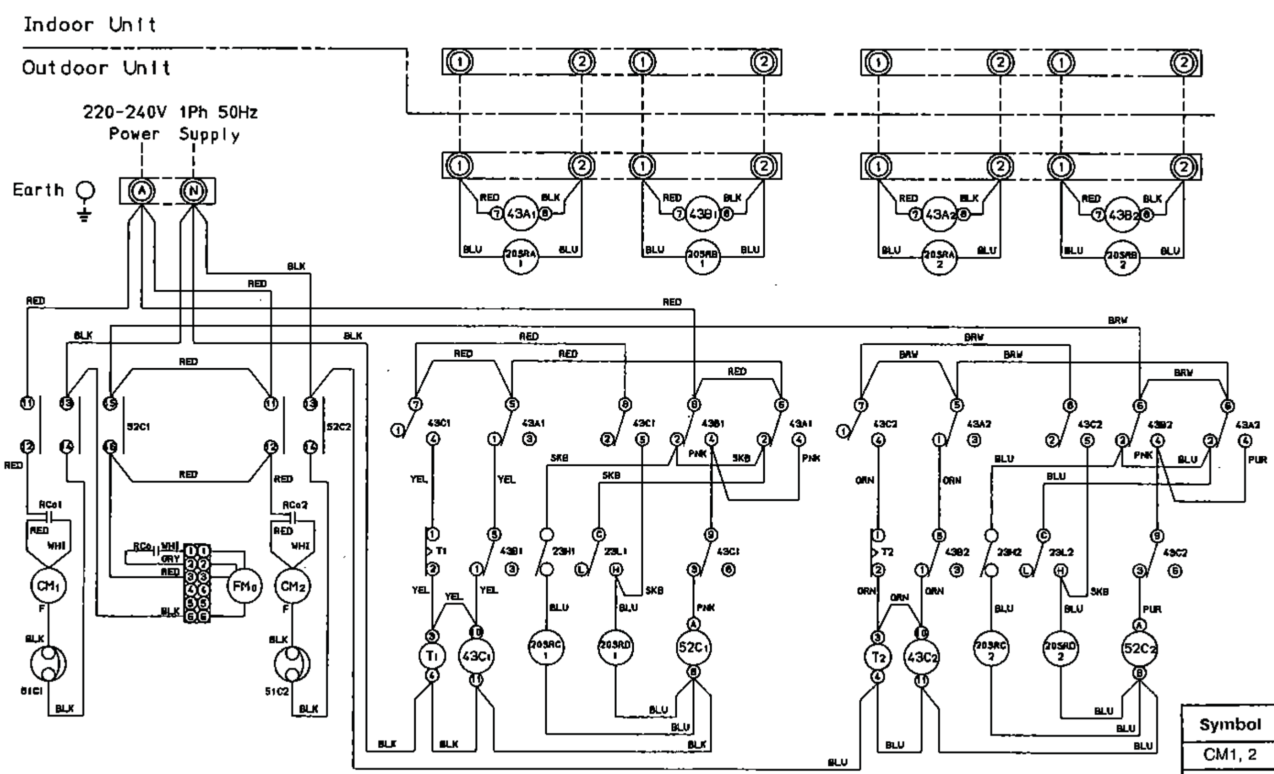
Remark: Be sure to refer to the service data file No. 300-900 for the indoor unit RAS-09UKR to be connected.

2. CONSTRUCTION VIEWS

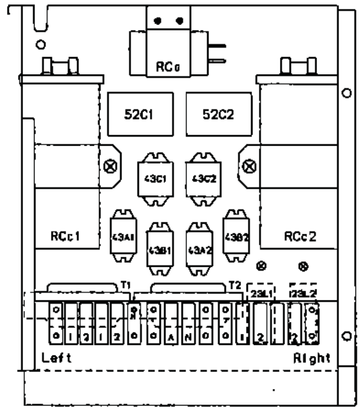
Outdoor Unit
RAV-M240A



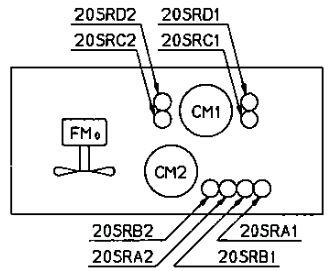
3. WIRING DIAGRAM



Parts Position
(Electrical Parts Box)



(Fan Motor. Compressor. Coil)



◎ shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

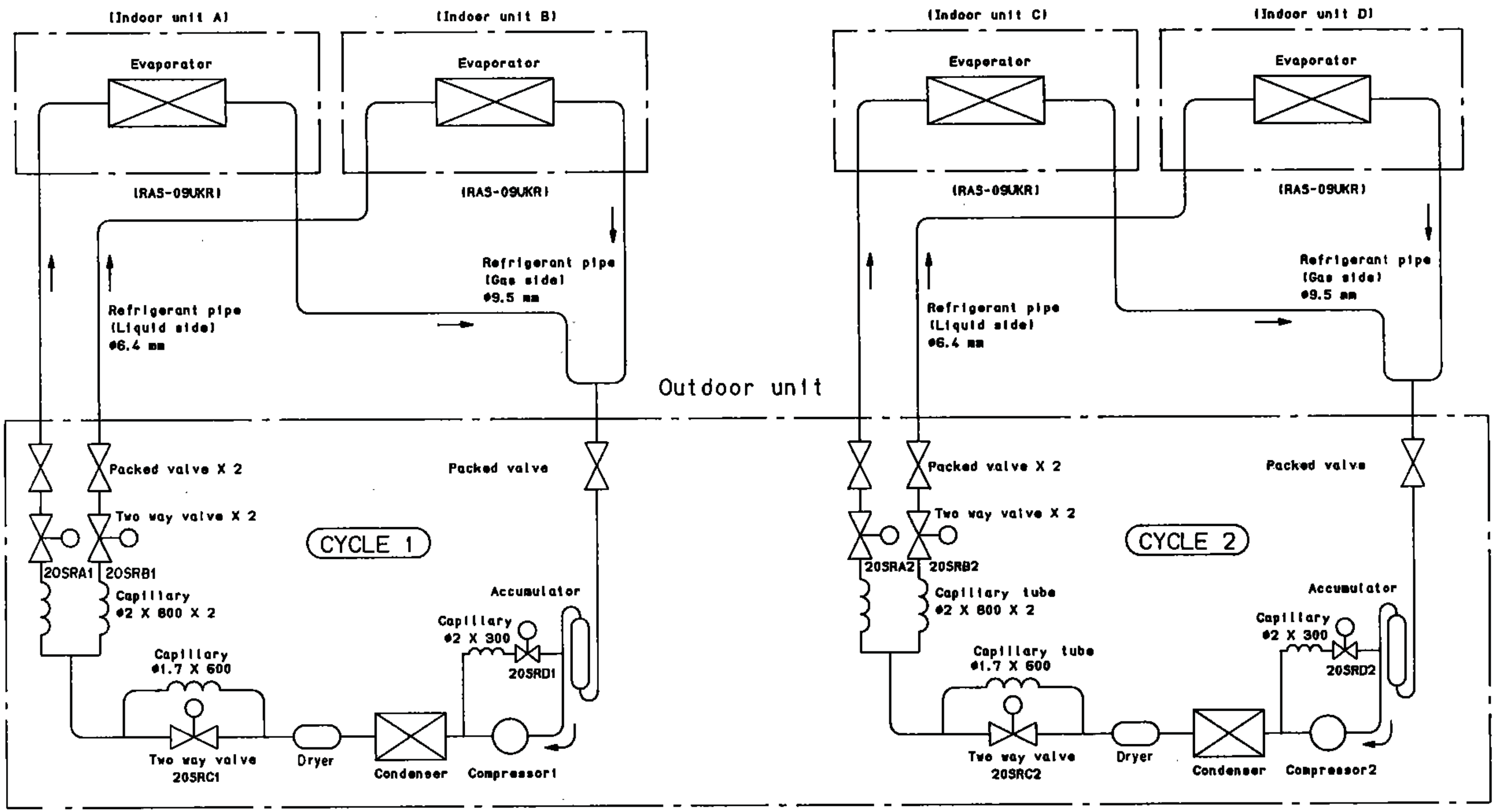
Don't operate the units with the magnetic contactor pushed.

Symbol	Name	Model name
CM1, 2	Compressor	PH160X2-4L
FMo	Fan Motor	STF-200-63B
RCc1, 2	Running Capacitor (Compressor)	MT-40MP356W
RCo	Running Capacitor (Fan Motor)	EAG45M355UF1
51C1, 2	Overload Relay	MRA99420
52C1, 2	Magnetic Contactor	VC20FA 3P AC240V
43A1, 2 43B1, 2	Relay	LY2F-AC220/240V
43C1, 2	Relay	LY3F-AC220/240V
20SRA1, 2 20SRB1, 2 20SRC1, 2 20SRD1, 2	Solenoid Coil	NEVAC240V
T1, 2	Thermal Timer	230-B-180-4BFP
23L1, 2	Thermostat	ATB-T306

4. SPECIFICATIONS OF ELECTRICAL PARTS

PARTS NAME	TYPE	SPECIFICATIONS
Compressor 1, 2	PH160X2-4L	Output 1.1kW, 2 pole, 1ϕ, 220-240V, 50Hz Winding resistance: Main coil 2.2Ω, Aux. coil 3.5Ω
Fan motor	STF-200-63B	Output 63W, 6 pole, 1ϕ, 220-240V, 50Hz
Magnetic contactor	VC20FA	AC240V
Capacitor	MT-40MP356W	For compressor 1, 2 : PH160X2-4L, AC400V, 35μF
	EAG45M355UF1	For fan motor, AC450V, 3.5μF
Solenoid coil (2-way valve)	NEV AC240	AC220-240V
Relay	LY2F	Coil 220-240V, 50/60Hz
	LY3F	Coil 220-240V, 50/60Hz
Thermal timer	230-B-180-4BFP	AC220-240V, 3-min. (Delay)

5. REFRIGERANT PIPING DIAGRAM
RAV-M240A



REFRIGERANT PIPING

Permissible Piping Length and Head

Model	Connectable indoor unit number	Permissible piping length	Permissible difference pipe lengths	Permissible piping head (H)	Between units A and B piping head (Δh)	Remarks
RAS-09UKR	4	15m ($\ell_1 + \ell_2$) 15m ($\ell_3 + \ell_4$)	8m ($\ell_1 - \ell_2$) 8m ($\ell_3 - \ell_4$)	5m	1m	Fig. 1

The minimum inter-unit refrigerant piping length shall be 2m.

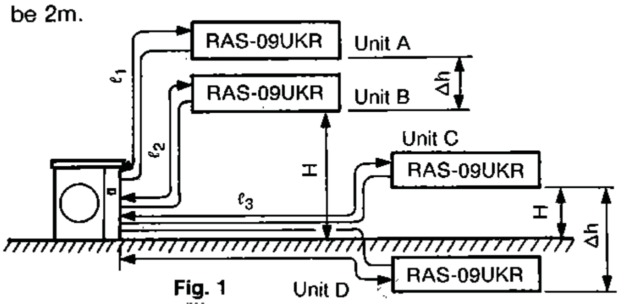


Fig. 1

Limit the number of bends in the refrigerant piping to 10 or less.

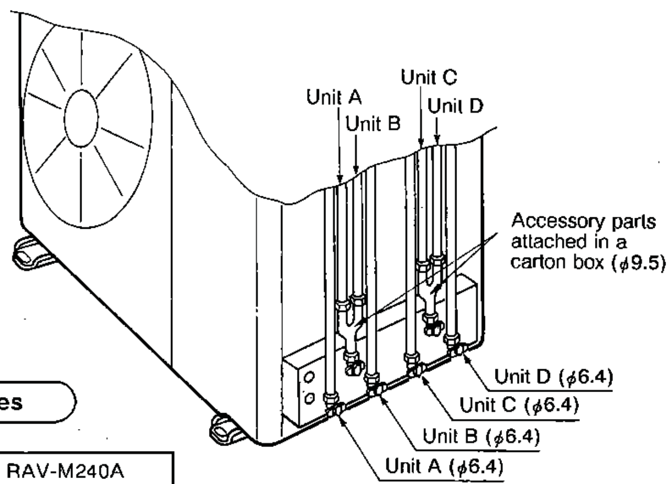
Piping Material and Sizes

Piping material		Phosphate deoxidized copper seamless pipes for air conditioners
Model		RAS-09UKR
Piping size (mm)	Larger	9.5
	Smaller	6.4

Air Purging

- Subject the refrigerant tube of outdoor and indoor units to air purge with a vacuum pump.
- Do not carry out this air purge by using the refrigerant filled in the outdoor unit.
- To handle valves, a 5mm hexagon wrench is needed.

Refrigerant Pipe Connecting Position



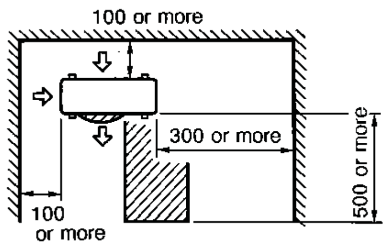
Additional Refrigerant Quantities

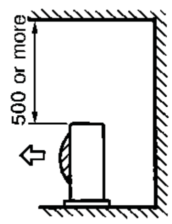
Model	RAV-M240A
Addition per meter	No need

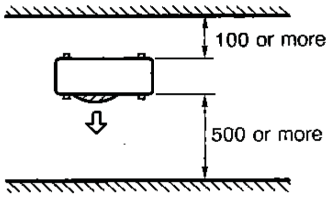
6. UNIT INSTALLATION

Service Space

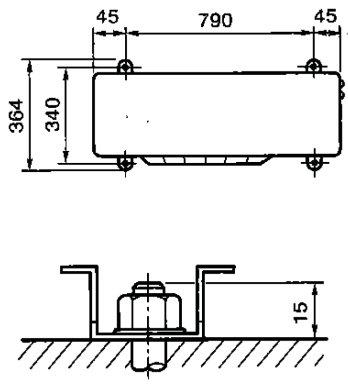
Ensure that there is sufficient space around the outdoor unit for installation and servicing.







- Do not install in a place that can increase the vibration and amplify the noise level of the units.
- Be sure to fix the outdoor unit with four (4) M10 anchor bolts according to foundation drawings below.



7. OPERATIONAL MATRIX OF ELECTRICAL PARTS

RAV-M240A

Parts / Operating indoor unit	Indoor unit PC board relay K7 (A)	K7 (B)	K7 (C)	K7 (D)	52C1	52C2	FMO	CM1	CM2	43A1 (A)	43B1 (B)	43A2 (C)	43B2 (D)	43C1	43C2	T1	T2	20SRA1 (Cycle 1)	20SRB1	20SRC1	20SRD1	20SRA2 (Cycle 2)	20SRB2	20SRC2	20SRD2
M240A Power ON No indoor unit operating	×	×	×	×	×	×	×	×	×	×	×	×	×	○	○	○	○	×	×	×	○	×	×	×	○
A	○	×	×	×	○	×	○	○	×	○	×	×	×	×	○	×	○	○	×	△	▽	×	×	×	○
B	×	○	×	×	○	×	○	○	×	×	○	×	×	×	○	×	○	×	○	△	▽	×	×	×	○
C	×	×	○	×	×	○	○	×	○	×	×	○	×	○	×	○	×	×	×	×	○	○	×	△	▽
D	×	×	×	○	×	○	○	×	○	×	×	×	○	○	×	○	×	×	×	×	○	×	○	△	▽
A B	○	○	×	×	○	×	○	○	×	○	○	×	×	×	○	×	○	○	○	×	×	×	×	×	○
A C	○	×	○	×	○	○	○	○	○	○	×	○	×	×	×	×	×	○	×	△	▽	○	×	△	▽
A D	○	×	×	○	○	○	○	○	○	○	×	×	○	×	×	×	×	○	×	△	▽	×	○	△	▽
B C	×	○	○	×	○	○	○	○	○	×	○	○	×	×	×	×	×	×	○	△	▽	○	×	△	▽
B D	×	○	×	○	○	○	○	○	○	×	○	×	○	×	×	×	×	×	○	△	▽	×	○	△	▽
C D	×	×	○	○	×	○	○	×	○	×	×	○	○	○	×	○	×	×	×	×	○	○	○	×	×
A B C	○	○	○	×	○	○	○	○	○	○	○	○	×	×	×	×	×	○	○	×	×	○	×	△	▽
B C D	×	○	○	○	○	○	○	○	○	×	○	○	○	×	×	×	×	×	○	△	▽	○	○	×	×
A C D	○	×	○	○	○	○	○	○	○	○	×	○	○	×	×	×	×	○	×	△	▽	○	○	×	×
A B D	○	○	×	○	○	○	○	○	○	○	○	×	○	×	×	×	×	○	○	×	×	×	○	△	▽
A B C D	○	○	○	○	○	○	○	○	○	○	○	○	○	×	×	×	×	○	○	×	×	○	○	×	×

K7 : Indoor unit cooling operation relay output
52C1, 52C2 : Compressor magnetic contactor
FMo : Outdoor unit fan motor
CM1, CM2 : Compressor
43A1,2 : Indoor unit cooling operation relay
43B1,2 : Indoor unit cooling operation relay
43C1,2 : Start delay relay
T1,2 : Thermal timer

20SR A1,2 : Indoor unit changeover 2-way valve solenoid coil
20SR B1,2 : Indoor unit changeover 2-way valve solenoid coil
20SR C1,2 : Compensation 2-way valve solenoid coil, OFF at the time of both indoor unit in a cycle in operation
20SR D1,2 : 2-way valve solenoid coil for bypass
○ : ON
× : OFF
△ : Discharge thermo switch ($23H_{1,2}$) ON at high temp. (OFF at low temp.)
▽ : Suction thermo switch ($23L_{1,2}$) ON at low temp. (OFF at high temp.)

8. TROUBLESHOOTING CHART FOR RAS-09UKR/RAV-M240A

Troubleshooting procedures	
1	Following the details of "What to be prechecked first", make sure of the basic items.
2	When there is not trouble corresponding to the items in "1" above, check in detail the faulty parts following "How to judge faulty parts by symptoms" later.

1. WHAT TO BE PRECHECKED FIRST:

1.1 Power voltage

The power voltage must be from AC 198V to 264V. If the power voltage is not within this range, the air-conditioner may not work normally.

1.2 Incorrect cable connection between indoor and outdoor units.

The indoor unit is connected to the outdoor unit with 2 cables. Make certain that the terminals of indoor and outdoor connectors have been connected properly by the same numbers. If not connected as specified, the outdoor unit won't operate normally.

1.3 Operations not regarded as failure (program operation)

In terms of the control of air-conditioner, the operations shown in table below are made as a program operation incorporated in a micro-processor. If a claim is made about the operation, check it corresponds to the contents in the table. If it does, it is an indispensable operation for the control and maintenance of the air-conditioner but not a failure of the units.

- **Operations which are not deemed trouble**

No.	Operation of Air-Conditioner	Description
1	When the POWER plug socket of the indoor unit is inserted, the operation lamp flashes.	The operation lamp flashes, indicating that power is turned on. If this happens, press the ON/OFF button once, and flash will stop. Power failure also causes the same lamp to flash.
2	Room temperature is in the range under which the compressor is turned on, but the compressor will not start.	The compressor will not start while the compressor restart prevention timer (three-minute timer) is actuated. This applies also when power is turned on.
3	Fan speed remains unchanged when the fan speed button is operated in the dry mode.	Fan speed is fixed at Low in the dry mode.
4	Room temperature is in the range under which the compressor is turned off, but the compressor will not stop.	The compressor will not stop while the compressor on-hold timer (two-minute timer) is actuated.
5	The compressor will not switch on or off even when the thermostat control is operated in the dry mode.	In the dry mode, the compressor goes on and off at regular intervals, independent of the thermostat control.

2. PRIMARY JUDGEMENT OF TROUBLE SOURCES

2.1 Role of indoor unit controller

The indoor unit controller receives the operation commands from the remote controller and assumes the following functions.

- Measurement of the draft air temperature of the indoor heat exchanger by using the temperature sensor (TA).
- Louver motor control
- Control of the indoor fan motor operation
- Control of the LED display
- Control of the outdoor unit compressor and the outdoor fan motor

2.1.1 Judgement from defective operation or abnormal operation

NO.	SYSTEM	CHECK		PRIMARY JUDGEMENT
1.	No reaction on remote control operation	Turn off the power once, turn it on again and try to operate the remote control again.	① Remote control is not possible.	The Indoor part (including the remote control) is defective.
			② Remote control is possible.	O.K.
2.	The outdoor fan does not rotate	① The compressor operates.		The outdoor part is defective (outdoor fan motor)
		② The compressor does not operate.		The indoor part is defective. The outdoor part is defective.

2.1.2 Self-diagnosis with remote controller

With the indoor unit controller, self-diagnosis of protective circuit action can be done by turning the remote controller operation into service mode, operating the remote controller, observing the remote controller indicators and checking whether ON lamp flashes (5Hz).

1. How to select remote controller operation mode

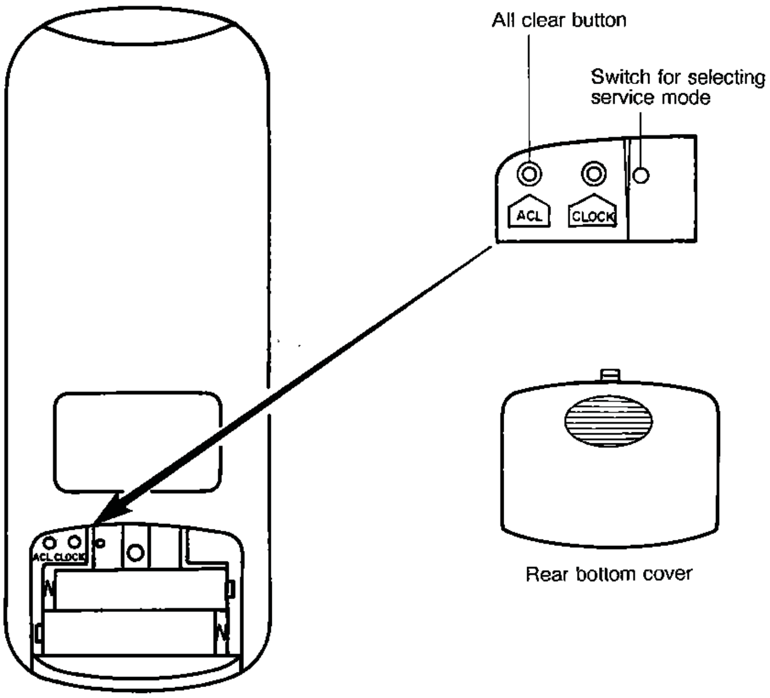
(1) Selecting service mode
Push the switch button provided on rear bottom of the wireless remote controller with a tip of pencil for more than 3 seconds. Make sure the setting temperature "   " is displayed on the display and other display is turned off.

(2) Selecting ordinary mode
Push the all clear button (ACL) on the rear bottom of the wireless remote controller with a tip of pencil for more than 3 seconds. Make sure the operation mode display, fan speed display, clock display and setting temperature display are turned on and " : " of the clock display is blinking.

2. Cautions when doing service

(1) After completion of servicing, always push the all clear (ACL) button to return the operation mode to the normal mode.

(2) After completion of servicing by the check code, turn off the power once and then turn on the power to reset memorized contents of the microprocessor to the initial status.



(Rear bottom of remote controller)

3. Self-diagnosis by check codes

(1) The self-diagnosis by the check codes is conducted under the block displays of item B-E.

(2) Remote control key operation under the service mode is conducted by ON/OFF or TEMP.
The remote control display by each key operation is varied as shown below. Two digit number is displayed in a hexadecimal number.

Operating key	Indication after operation
ON/OFF	" 00 "
TEMP. △ (Up)	1 is added to data before operation. (Example) " 02 " → " 03 "
TEMP. ▽ (Down)	1 is subtracted from data before operation. (Example) " 02 " → " 01 "
"AUTO" LOUVER	10 is subtracted from data before operation. (Example) " 02 " → " 12 "
"SET" LOUVER	Data before operation is directly transferred. (Example) " 02 " → " 02 "

(3) The self-diagnosis by the check codes is conducted with procedures shown below.
1) Enter the service mode and make sure the off timer display of the remote control shows " 00 ".
2) Operate the "ON/OFF" key and make sure the timer lamp on the display section is blinking (5Hz).
3) At the same time, also make sure the operation lamp is also blinking. This shows that the protection circuit on the indoor PC board is working.
4) Operate the TEMP. △ key and make sure the remote control display shows " 01 " and blinking of the operation lamp. If the operation lamp is blinking, it shows the protection circuits for power line and serial signal system is working.
5) In the same way, operate the TEMP. △ key so the display is incremented by +1 to continue checks by the self-diagnosis as shown in the table below. From " 00 " up to " 03 " check operations of protection circuits for each block, and " 04 " to " 1F " check operations of the typical protection circuits.

3.1 Display of abnormalities and judgement of the abnormal spots

The indoor unit observes the operation condition of the air conditioner and displays the contents of the self-diagnosis as block displays on the display panel of the indoor unit.

	BLOCK DISPLAY	CHECK CODE	SELF-DIAGNOSIS	CHECK CODE
A	Operation Display Flashing (1 Hz)	–	Power Failure (When power is on)	–
B	Operation Display Flashing (5 Hz)	00	Temperature Sensor (TA) Short/Break	0C
C	Operation Display Flashing (5 Hz)	00	Heat Exchanger Sensor (TC) Short/Break	0d
D	Operation Display Flashing (5 Hz)	00	Indoor Fan Lock Abnormality of Indoor Fan, IC03 Short/Break	11
E	Operation and Timer Display Flashing (5 Hz)	01	Thermal Fuse is Blown (Indoor Fan Motor is Overheat)	04

3. TROUBLESHOOTING FLOWCHARTS

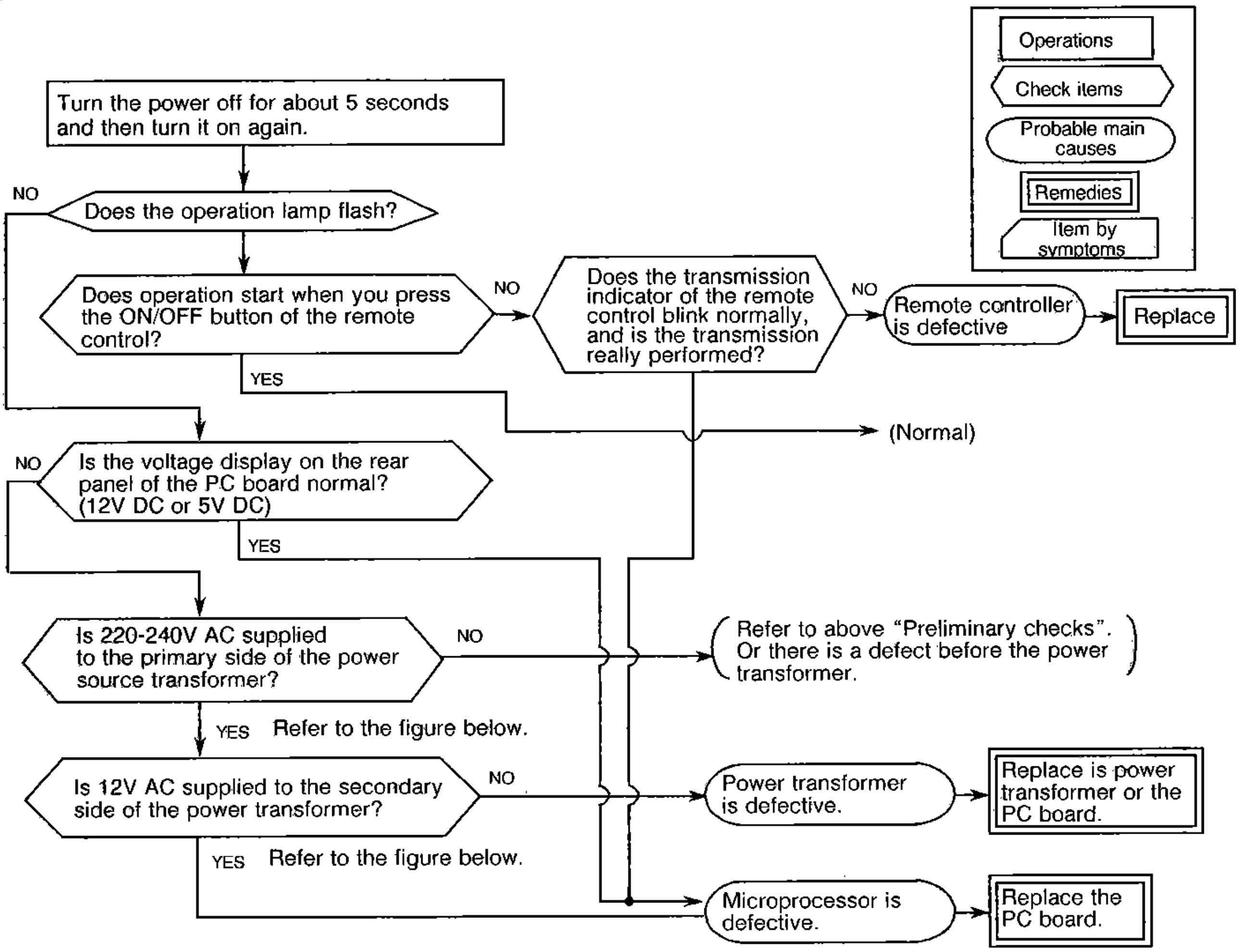
3.1 Power cannot be turned on (No operation at all)

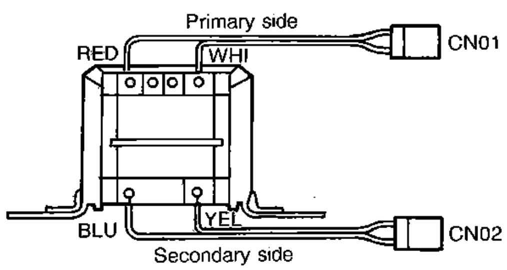
<Preliminary checks>

① Is the supply voltage normal?

② Is the connection to the AC output O.K.?

③ Are the connection of the primary side and the secondary side of the power transformer inserted into the PC board?





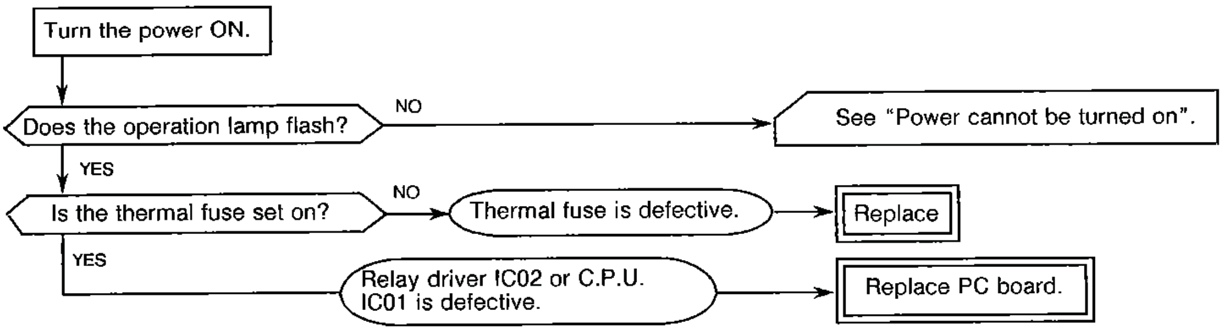
Power transformer connection diagram.

Color Identification

WHI:	WHITE
RED:	RED
YEL:	YELLOW
BLU:	BLUE

3.2 Power relay RY7 does not operate.

Louver is not controlled automatically.



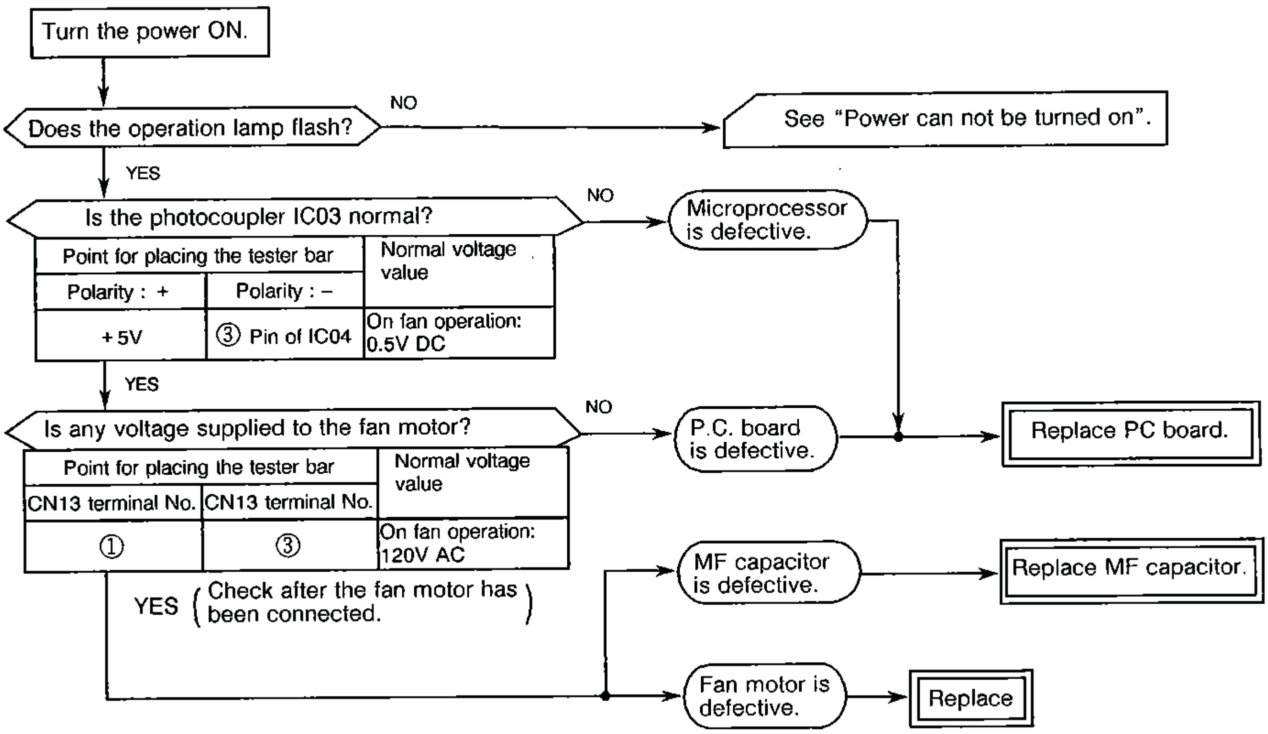
- When wiring to the thermal fuse has been broken, the timer lamp and operation lamp will flash with 5 Hz.

3.3 Only the indoor fan does not operate

<Preliminary checks>

Does it neither work in COOL or FAN ONLY operation?

<Checking procedure>



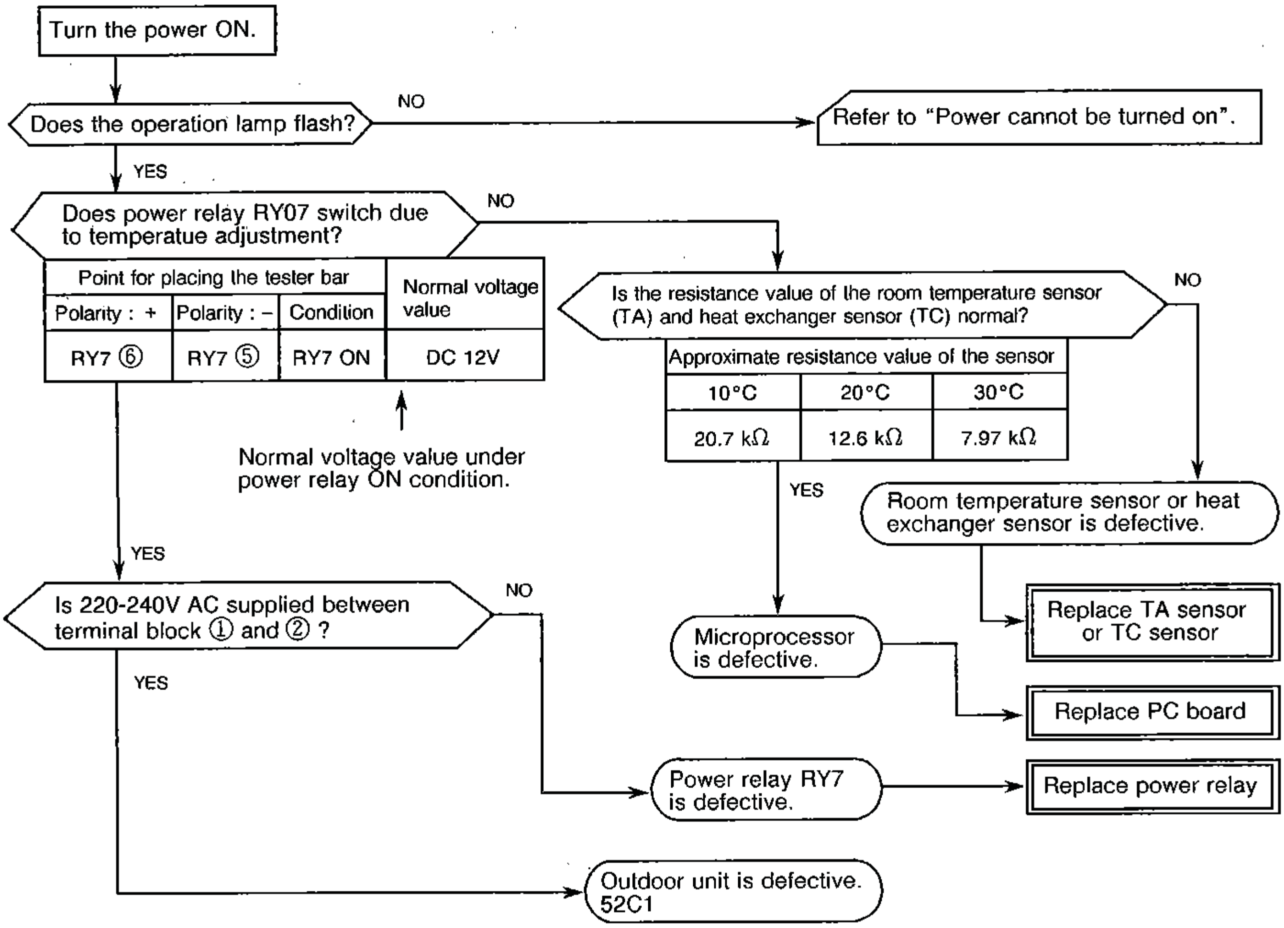
3.4 Compressor does not operate (Indoor fan also does not turn.)

< Preliminary checks >

① Is the temperature set on the remote control higher than the room temperature in cool operation?

② Is contact of the inter-unit wiring O.K.?

< Checking procedure >

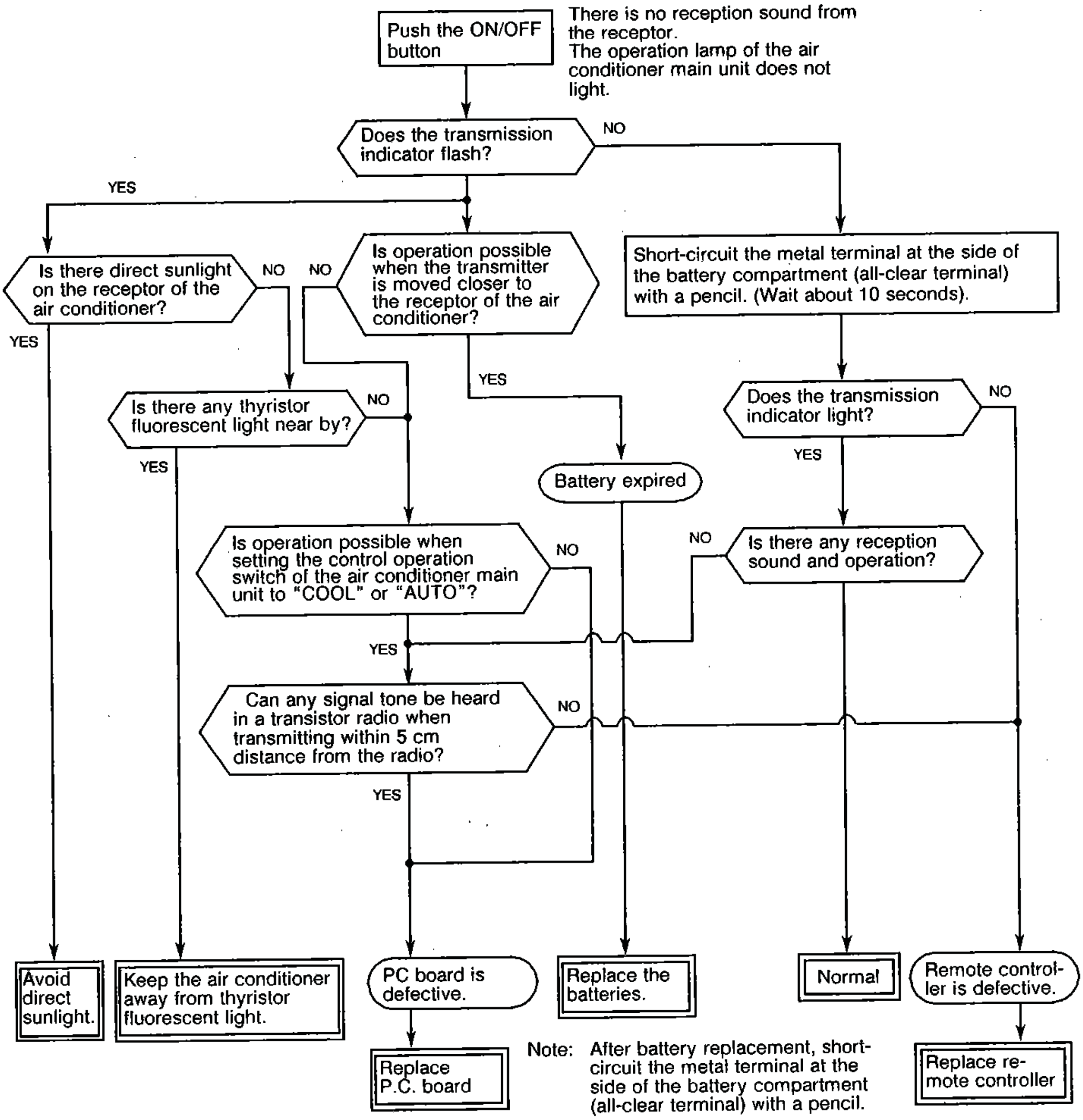


4. TEST POINTS ON THE PC BOARD AND LIST OF VOLTAGE VALUES

The test points (TP) are indicated on the rear of the PC board.
The voltage values on the test points for defect diagnosis items are listed below.

No.	Defect diagnosis item	Point for placing the tester bar			Normal voltage value
		Polarity (+)	Polarity (−)	Condition	
①	Does the power relay (RY7) switch?	RY7 ⑥	RY7 ⑤	When relay RY7 is ON.	12V DC (When relay is ON.)
②	Is the voltage of the secondary side of the transformer O.K.?	TP3	TP5	–	No load : 14 ± 2V With load: 12 ± 2V (Primary voltage is about 230V)
③	Is the voltage of the primary side of the transformer O.K.?	TP1	TP2	–	220-240V AC
④	Are both PTH terminals 5V AC or less?	Either end of PTH		–	5V AC or less : normal 5V AC or more: abnormal

5. HOW TO CHECK THE REMOTE CONTROLLER (INCLUDING THE INDOOR PC BOARD)



5.1 HOW TO CHECK THE PC BOARD

5.1.1 Operating precautions

① When removing the front panel or the PC board, be sure to disconnect the power plug from the AC outlet.
② When removing the PC board, hold the edge of the PC board and do not apply force to the parts.
③ When connecting or disconnecting the connectors on the PC board, hold the whole housing. Do not pull at the lead wire.

5.1.2 Inspection procedures

① When a PC board is judged to be defective, check for disconnection, burning, or discoloration of the copper foil pattern on this PC board.
② The PC board consists of the following 4 parts:

1. Main PC board part................ power relay, indoor fan motor drive circuit and control circuit, C.P.U. and peripheral circuits, buzzer drive circuit and buzzer.
2. Light receiving unit................ light receiving circuit
3. Display .... LED
4. Switch PC board .... wireless-control, temporary switch

Check the defects of the PC board following the list below.
(Referring to Table-4)

5.1.3 Checking procedure

No.	Procedure	Check point (Symptom)	Trouble cause
①	Disconnect the power plug from the AC outlet and remove the PC board assembly from the electronic parts base. Remove the flat cable from the terminal plate.	Is the fuse blown?	① Application of shock voltage ② Short-circuit of the indoor fan motor
②	Remove the motor connector from "MOTOR". Turn the power ON. If the operation lamp flashes (0.5 sec. ON, 0.5 sec. OFF), steps ①–③ in the right column are not necessary.	Check power supply voltage. ① Between CN01 ① and ③ (220/240V AC) ② Between CN02 ① and ③ (12V AC) ③ Between Q01 emitter and GND (5V DC) ④ Between Q03 emitter and GND (12V DC) ⑤ Between CN11 ① pin and GND (12V DC)	① Defective power cord, power switch, fuse or line filter, or wrong wiring ② Defective power transformer ③ Defective power circuit or short-circuited load ④ Same as ③ ⑤ Thermal fuse operation
③	Push the ON/OFF button once to set in operation mode. (Do not set to the fan only or on-timer mode.)	Check power supply voltage. ① Power relay coil voltage (12V DC) (RY7 ⑤,⑥) ② Between SL connectors 1 and 2	① Relay coil cable is broken, relay driver (IC181) is defective. ② Relay contact is defective, SL connector is defective.
④	Start operation by using the anti-restart timer.	① All LEDs of the operation lamp, the timer lamp and the FAN ONLY lamp light up. ② After 3 seconds, normal display does not appear.	Display is defective or defect in the 4P housing assembly.

No.	Procedure	Check point (Symptom)	Trouble cause
⑤	Push the ON/OFF button once to set in operation mode. 1. Setting the anti-restart timer 2. Cooling operation 3. Fan speed : AUTO 4. Set the temperature sufficiently lower than the room temperature. 5. Continuous operation	① The compressor does not operate. ② The operation lamp flashes.	① The temperature of the Indoor heat exchange unit is extremely low. ② Defective control PC board.
⑥	Connect the motor connector to "MOTOR" and turn the power ON. Start operation as follows: 1. Set the operation mode to "FAN ONLY". 2. Set the fan speed to "HIGH". 3. Continuous operation	① There is a voltage of 120V or more between the red and black motor connector leads. ② The motor does not rotate. (But the key operation of the remote control is accepted.) ③ Motor rotates but vibrates hard.	① Indoor fan motor is defective. ② Contact of the motor connector is defective. ③ Main PC board is defective.

Approximate value of the sensor (thermistor) resistance (TA, TC)

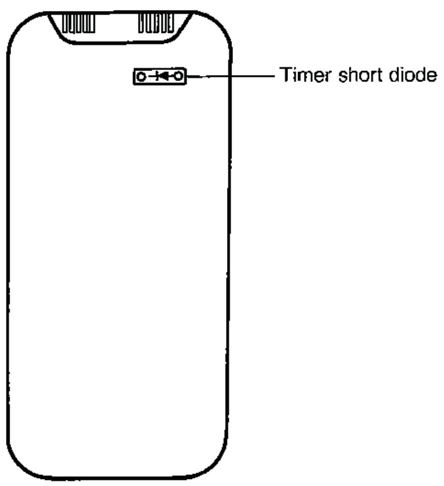
(≒ kΩ)

Sensor \ Temperature	0°C	10°C	20°C	30°C
Thermo Sensor	35.8	20.7	12.6	7.97

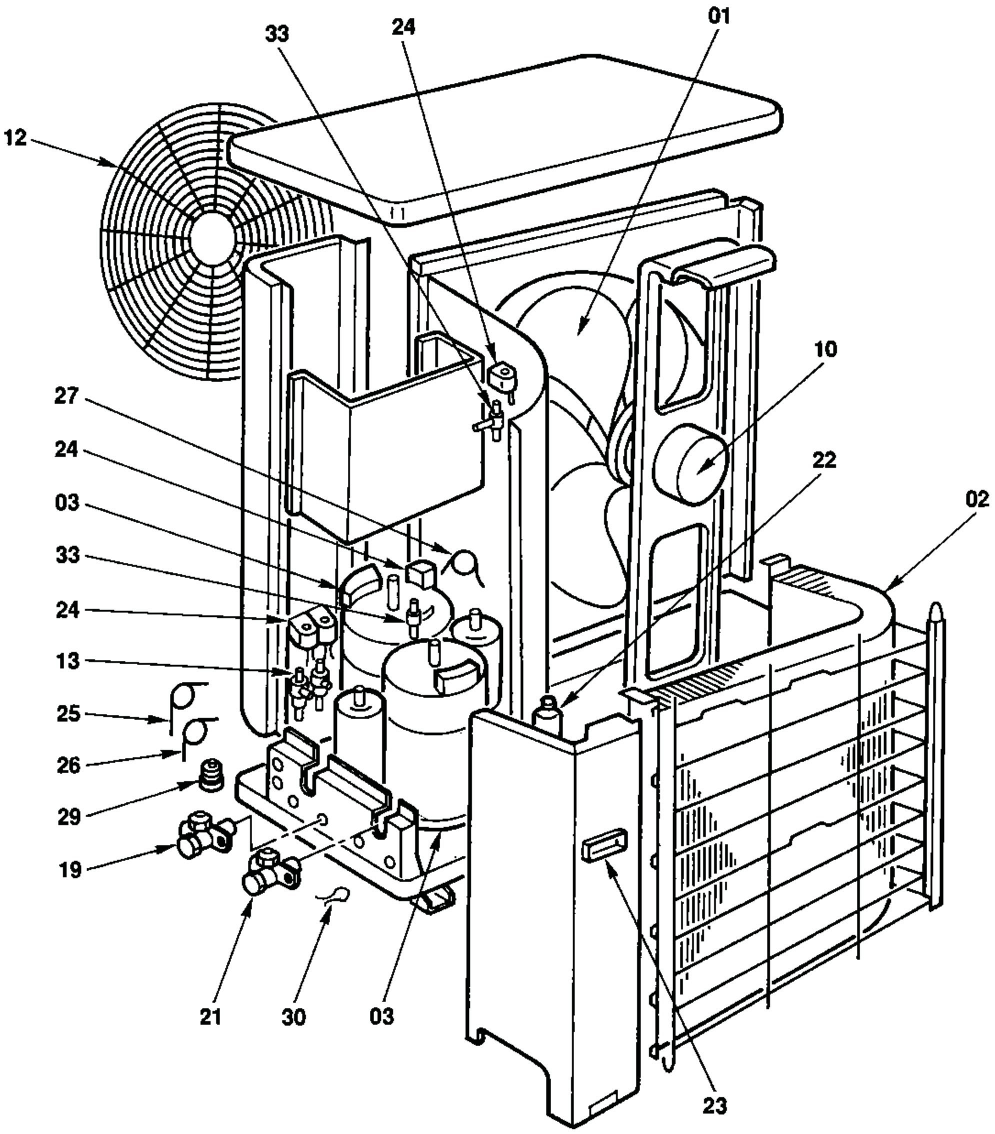
5.2 How to reduce the operation time of the anti-restart timer

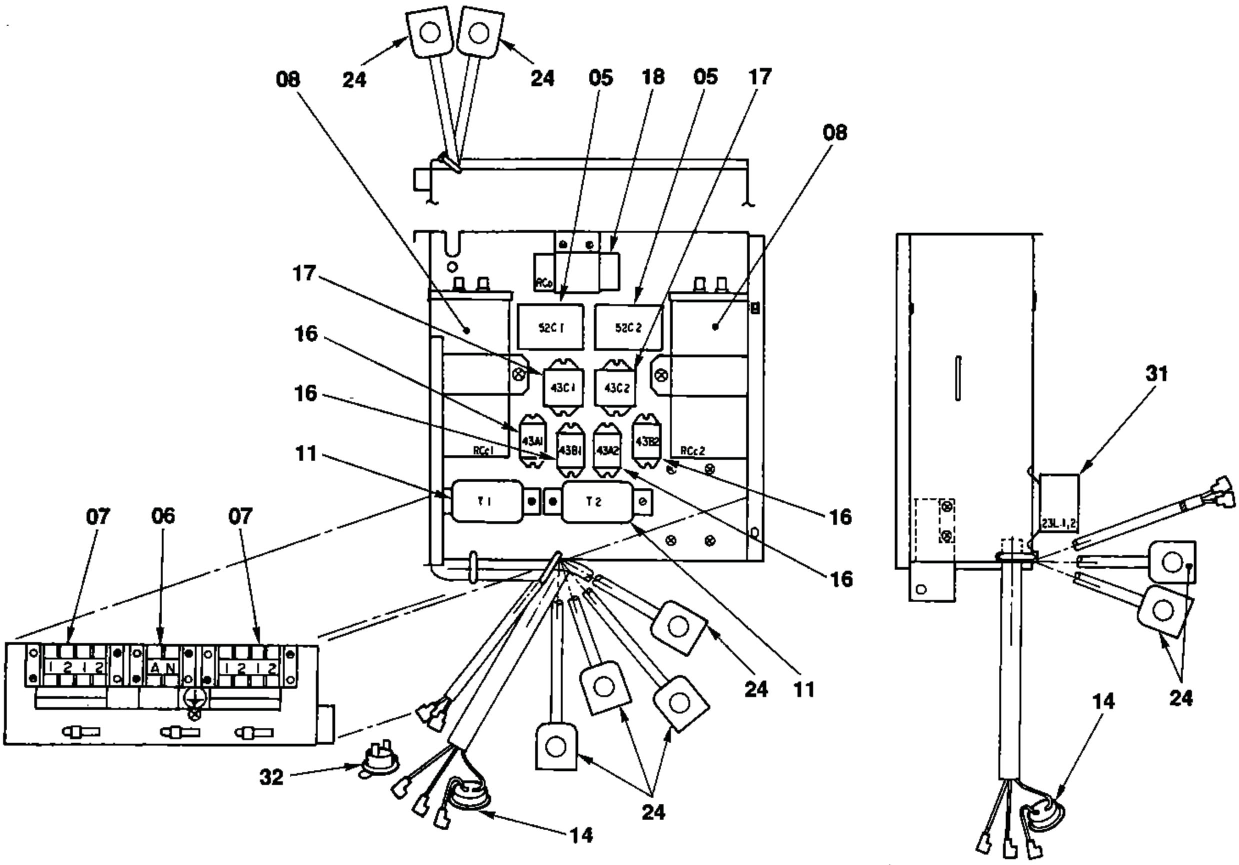
- Drill 2 holes on the rear of the wireless remote control unit.
  Attach the diode (IS1555 or equivalent) to the rivet inside the unit.
- Push the ON/OFF button to start operation with the diode attached.
  During operation, push the transmission/check button while the diode is attached.

Wireless remote controller



9. EXPLODED VIEWS AND PARTS LIST





Location No.	Part No.	Description
01	43120168	Fan, Propeller
02	43143701	Condenser
03	43041728	Compressor, AC, 220/240V, 50Hz PH160X2-4L
05	43154139	Magnetic-Switch
06	43060774	Terminal, Block
07	43160466	Terminal, Block, 4P
08	43055354	Capacitor, Plastic Film, 35MFD, 400V
10	43121636	Motor, Fan, STF-200-63B
11	43051259	Timer, Thermal
12	43191315	Guard-Fan
13	43046270	2-Way Valve
14	43054380	Relay, Overload
16	43154141	Relay, LY2F-L
17	43154135	Relay

Location No.	Part No.	Description
18	43155146	Capacitor, Electrolytic
19	43046229	Packed Valve, 3/8
21	43046228	Packed Valve, 6.35
22	43145103	Dryer
23	43119390	Hanger
24	43146443	Solenoid Coil
25	43047492	Capillary Tube 1.7 Dia
26	43047491	Capillary Tube 1.5 Dia
27	43047527	Capillary Tube 2.0 Dia
29	43049625	Cushion, Rubber
30	43069988	Holder, OL-Relay
31	43150202	Thermostat
32	43150220	Bimetal Thermostat
33	43046151	2-Way Valve